

Zhongyin Wang

+1 (217) 607-7987 | zw92@illinois.edu | [GitHub: joeywang02](#) | [LinkedIn](#)

Education

University of Illinois Urbana-Champaign

Urbana, IL, USA

Master of Science in Information Management (GPA: 4.0/4.0)

Aug. 2025 – May 2027

- **Coursework:** CS 411 Database Systems, IS 515 Information Modeling, CS 427 Software Engineering I, IS 507 Statistical Models and Information, IS 504 Sociotechnical Info Systems.

Zhejiang University

Hangzhou, China

Bachelor in Civil Engineering (Smart Transportation & Systems) (GPA: 3.6/4.0)

Sept. 2020 – July 2024

- **Awards:** Zhejiang University Excellent Graduate, Structural Design Competition Second Prize.
- **Coursework:** Data Structures and Algorithm Analysis, Fundamentals of CS, C Programming.

Projects & Academic Research

Nexus | AI-powered Education Planner

Mar. 2026 – Present

- Developed an intelligent scheduling system that automates the parsing of heterogeneous syllabus data (PDF/JSON) from **Canvas** and Coursera by integrating **LLM APIs** for semantic extraction.
- Engineered a conflict-detection algorithm to generate optimized study plans, utilizing **Redis caching** for course metadata to ensure low-latency API responses.
- Implemented **Spring AI** to design a centralized storage standard for academic tasks, focusing on personalized learning pathways to reduce students' cognitive load.

Enterprise Compliance & Rewards Platform

Nov. 2025 – Jan. 2026

- Developed a **graph-based data model** using **Neo4j** to model hierarchical compliance policies and reward rules, enabling efficient knowledge representation and semantic path traversal.
- Optimized complex **Cypher** queries for deep-link analysis across heterogeneous entities (Employees, Awards, Policies), improving logic retrieval speed compared to traditional SQL joins.
- Implemented a **Redis Cache-Aside pattern** to mitigate high-concurrency bottlenecks, reducing MySQL database load by 30% while ensuring ACID transaction integrity.
- Designed resilient RESTful APIs with comprehensive JUnit/Mockito coverage; containerized the system with Docker to ensure environment parity.

LLM Based Multi-Table Research on Credit Data

Sept. 2024 – Dec. 2024

Research Assistant / Advisor: Prof. Yang Yang (Zhejiang University)

- Engineered a cross-table learning framework leveraging **Large Language Models (LLMs)** to perform semantic extraction and scoring on heterogeneous financial datasets.
- Replicated and benchmarked mainstream models based on "Cross-Table Masked Pretraining" to establish reliable baselines for web data mining tasks.
- Designed and implemented a dual-scale model structure supporting both **row-based sampling** for fine-grained regression and **table-level sampling** for distribution-aware classification.
- Optimized fine-tuning pipelines to improve the semantic alignment of LLMs with structured features, enhancing credit risk evaluation accuracy.

Transformer-based Urban Road Network Design (Generative AI)

Dec. 2023 – June 2024

Bachelor's Thesis / Advisor: Prof. Hongsheng Qi (Zhejiang University)

- Developed a generative AI pipeline using **Transformer** and **Graph Neural Networks (GNN)** to automate the design of urban road networks from zero-to-one.
- Processed large-scale geospatial data via OSM APIs, converting raw geographic nodes into optimized **graph-based topologies** to evaluate connectivity and network resilience.
- Formalized structural constraints into the generative process, bridging the gap between deep learning architectures and real-world civil engineering requirements.
- Validated model performance by comparing generated layouts with existing traffic networks to support autonomous driving simulations.

Smart Tunnel Dynamic Emergency Response System

May 2022 – May 2023

Project Leader / Advisor: Prof. Ke Wu (Zhejiang University)

- Integrated tunnel fire dynamics models with multi-sensor data fusion to predict fire temperature fields and smoke spread in real-time.
- Leveraged **YOLO** algorithms and computer vision to automate fire source detection and scale estimation, reconstructing fire scenes based on physical dynamics equations.
- Developed a dynamic response platform using **Java**, enabling real-time emergency coordination and response for critical infrastructure safety.

Experience

- Baidu, Inc. | NLP Algorithm Engineer

Beijing, China | Apr. 2024 – June 2024

 - Engineered a modular benchmarking infrastructure in Python to evaluate **50+ LLMs**, supporting concurrent local and API-based deployments for large-scale model evaluations.
 - Conducted ROI assessments for Agent architectures including ReAct and Plan-and-Execute, analyzing TTFT and task success rates to support strategic infrastructure selection.
 - Developed automated pipelines to audit **SFT data quality**, identifying logic bottlenecks and model hallucinations.
- IDX Exchange | Data Analyst Intern

Boise, ID (Remote) | Mar. 2026 – Present

 - Built automated Python ETL pipelines to aggregate MLS transaction datasets; integrated FRED API for time-series analysis and engineered market intelligence metrics to support real estate decision-making.

Technical Skills

- Languages:** Python, Java, C/C++, SQL, JavaScript, Cypher (Neo4j)
- Frameworks:** Spring Boot, RESTful APIs, Spring AI, PyTorch, JUnit, Pytest
- Databases & Tools:** Neo4j, Redis, MySQL, PostgreSQL, Docker, Git, Linux
- AI Specialized:** LLM Prompt Engineering, Agent Orchestration, Knowledge Representation

Faculty References

- Jim Hahn** | Associate Professor, University Library at UIUC | jimhahn@illinois.edu
Reference Context: Instructor for IS 515 (Information Modeling) at UIUC
- Hongsheng Qi** | Associate Professor, Zhejiang University | qihongsheng@zju.edu.cn
Reference Context: Bachelor's Thesis Supervisor at Zhejiang University